TECHNICAL SPECIFICATION & ARCHITECTURAL DOSSIER

# PlacePrep Enterprise Platform

## Comprehensive Architectural Blueprint, Directory Tree, Operational Workflows & Database Schema Specification

**System Name:** PlacePrep Multi-Tenant Academic & Placement Suite
**Live Production URL:** https://place-prep-alpha.vercel.app/
**Source Code Repository:** https://github.com/Priyansh1867/PlacePrep
**Technology Stack:** React 19, Vite, Tailwind CSS, Spring Boot 3.5.16, Java 23, MySQL 8.0, Spring Data JPA, Hibernate, JWT
**Author / Engineering Lead:** Priyansh Jain & PlacePrep Engineering Team
**Version:** 2.4.0 (Enterprise Release with Geofencing & Multi-User Profile Sync)
**Classification:** Confidential Engineering Document
**Date of Issue:** October 2026

**EXECUTIVE SUMMARY:**
PlacePrep is an end-to-end, enterprise-grade educational technology and campus placement automation platform designed for engineering universities and technical institutions. The platform bridges the operational chasm between three distinct institutional stakeholders: **Students** preparing for competitive software engineering interviews and tracking academic credentials, **Faculty & Department Heads (HODs)** administering academic curriculums, monitoring class sessions, and enforcing strict geofenced classroom attendance, and **Training & Placement Officers (TPOs)** managing multi-round recruitment drives, company-specific candidate shortlisting, and real-time eligibility radar.

This technical dossier delineates the complete multi-tier software architecture, the authoritative codebase tree structure across frontend and backend modules, the algorithmic mechanisms underpinning core workflows (notably 100-meter GPS Geofencing with facial verification and multi-user profile synchronization), and the comprehensive relational database schema implemented in MySQL.

| Frontend Stack | Backend Engine | Database & Storage | Security & Compliance |
|---|---|---|---|
| React 19 + Vite | Spring Boot 3.5.16 | MySQL 8.0 Server | Spring Security 6 |
| Tailwind CSS v3 | Java 23 LTS runtime | Relational InnoDB engine | JWT Stateless Auth |
| Lucide Icons + Chart.js | Spring Data JPA + ORM | LocalStorage Cache Sync | 100m GPS Haversine |
| Monaco / CodeEditor | HikariCP Connection Pool | Normalized ER Schema | Role-Based Access (RBAC) |

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# 1. System Vision, Objectives & Core Personas

PlacePrep addresses systemic inefficiencies in traditional university placement cells and academic monitoring systems. Standard university administration relies on disparate spreadsheets, fragmented attendance registers, and disconnected online coding platforms. PlacePrep consolidates these operational facets into an integrated, reactive single-page architecture backed by an enterprise Spring Boot microservice.

| Institutional Persona | Role Key | Functional Scope & Capabilities |
|---|---|---|
| **Student** | `student` | Accesses personal profile, explores company-specific archives, practices in DSA Studio, runs ATS Resume audits, views enrolled subject attendance, and marks class attendance strictly within the 100m geofence radius. |
| **Faculty / Teacher** | `teacher / faculty` | Conducts speed class attendance marking, initiates daily classroom lecture sessions, verifies student attendance statistics, exports semester marksheet logs, and evaluates candidates. |
| **Head of Department** | `hod` | Monitors department-wide academic health, tracks faculty operations, evaluates active backlogs across cohorts, regulates curriculum governance, and identifies students below 75% attendance. |
| **TPO / Placement Cell** | `tpo` | Configures CRT sessions, schedules placement drives, creates company profiles, monitors student rounds progression, and manages candidate offer finalization. |
| **Institutional Management** | `management / admin` | Configures authoritative college coordinates (lat, lon, 100m radius), oversees campus-wide user directories, audits security logs, and manages system configurations. |

# 2. Complete Codebase Directory Tree

## Part A: Root Configuration, Tooling & Build Pipelines

```
PlacePrep (Project Root)/
|-- .env.example                                # Environment variables template for API endpoints
|-- package.json                                # NPM dependencies (React 19, Lucide, Tailwind, Vite)
|-- vite.config.js                              # Vite bundler build config with dev proxy to 8082
|-- tailwind.config.js                          # Tailwind CSS typography, color palette, and animations
|-- vercel.json                                 # Vercel deployment routes, headers, and SPA rewrites
|-- index.html                                  # Root HTML entry-point with viewport and favicon
|-- README.md                                   # Project onboarding, installation, and deployment docs
|-- PROGRESS.md                                 # Chronological implementation and feature changelog
|-- .oxlintrc.json                              # JavaScript / JSX linter configuration
|-- .github/                                    # CI/CD Workflows & modernization scripts
|   |-- modernize/java-upgrade/hooks/scripts/   # Automation tools & diagnostic hooks
```

## Part B: Frontend Single Page Application (src/)

```
src/                                            # FRONTEND REACT APPLICATION (SPA)
|-- App.jsx                                     # Root layout coordinator, authentication check, tab router
|-- main.jsx                                    # React DOM 19 createRoot client bootstrapping
|-- index.css                                   # Global CSS, Tailwind utilities & custom scrollbars
|
|-- components/                                 # REUSABLE FRONTEND UI & DOMAIN COMPONENTS
|   |-- AuthModal.jsx                           # Dual Login / Registration modal with role switching
|   |-- Navbar.jsx & Sidebar.jsx                # Navigation headers, active role indicators & mobile menu
|   |-- PortalModuleCards.jsx                   # Interactive dashboard navigation tiles
|   |-- ErrorBoundary.jsx                       # Component lifecycle error catcher
|   |-- BulkImportModal.jsx                     # CSV/Excel student roster ingestion
|   |
|   |-- academic/                               # ACADEMIC MODULES
|   |   |-- AcademicDashboardView.jsx           # Semester GPA, CGPA distribution & cohort summary
|   |   |-- BacklogManagementView.jsx           # Live backlog tracker & exam clearance monitoring
|   |   |-- StudentDirectoryView.jsx            # Searchable, filterable student roster with modal view
|   |   |-- StudentProfileModal.jsx             # Read-only academic candidate dossier
|   |
|   |-- attendance/                             # ATTENDANCE & GEOFENCING SUBSYSTEM
|   |   |-- StudentGpsAttendanceWidget.jsx      # Primary Student GPS (100m) + Face Recognition wizard
|   |   |-- DailyAttendancePortalView.jsx       # Faculty speed-marking portal & semester subject table
|   |   |-- CrtAttendancePortalView.jsx         # Campus Recruitment Training (CRT) session portal
|   |
|   |-- curriculum/                             # CURRICULUM & SYLLABUS GOVERNANCE
|   |   |-- CurriculumGovernanceView.jsx        # Department syllabus tree, credits, and coordinators
|   |
|   |-- dsa/                                    # DSA CODING & PRACTICE WORKSPACE
|   |   |-- DsaCodingWorkspace.jsx              # Interactive coding IDE, test cases, code execution
|   |   |-- DsaQuestionExplorer.jsx             # Searchable DSA question library (LeetCode-style)
|   |
|   |-- monitoring/                             # HOD FACULTY MONITORING
|   |   |-- FacultyOperationsMonitoringView.jsx # Faculty compliance radar, class conduction metrics
|   |
|   |-- profile/                                # MULTI-USER STUDENT PROFILE
|   |   |-- UserProfileModal.jsx                # Comprehensive 11-section candidate profile editor
|   |
|   |-- ui/                                     # DESIGN SYSTEM PRIMITIVES
|       |-- SearchBar.jsx, Badge.jsx, Button.jsx, Card.jsx, Dialog.jsx, Tabs.jsx, Toast.jsx
```

## Part C: Frontend Pages, Services & Backend Architecture

```
src/pages/                                      # TOP-LEVEL PAGE ROUTE VIEWS
|-- HomePage.jsx                                # Main student landing hub & quick-access tiles
|-- InstitutePortalPage.jsx                     # Master portal tab host (Academic, Attendance, ATS, DSA)
|-- PlacementPortalPage.jsx                     # TPO recruitment drive management dashboard
|-- ManagementPortalPage.jsx                    # College geofence coordinates configuration
|-- CompaniesPage.jsx & CompanyHubPage.jsx      # Placement companies directory & past interview records
|-- PlacementDrivesPage.jsx                     # Active hiring drives, schedules, and deadlines
|-- ResumeAtsCheckerPage.jsx                    # Automated resume analyzer, JD matcher & score report
|-- MockInterviewPage.jsx                       # AI-guided mock interview simulator
|-- SelectionWorkflowPage.jsx                   # Candidate round-by-round advancement pipeline
|-- ReportsExporterPage.jsx                     # Automated PDF/CSV institutional report generator
|
src/services/                                   # FRONTEND DATA ACCESS & CLIENT ENGINES
|-- api.js                                      # Centralized Axios/Fetch client with JWT interceptor
|-- storage.js                                  # Multi-user LocalStorage isolation & initial state
|-- attendanceStorage.js                        # Academic session & roster storage fallback
|-- crtAttendanceStorage.js                     # CRT attendance session storage fallback
|-- reportsExporterService.js                   # Client-side CSV/PDF generation utility
|
backend/src/main/java/com/placeprep/           # JAVA SPRING BOOT 3.5 ENTERPRISE SERVER
|-- PlacePrepApplication.java                   # Main application bootstrapper (@SpringBootApplication)
|
|-- config/                                     # CONFIGURATION & INFRASTRUCTURE
|   |-- SecurityConfig.java                     # Spring Security 6, JWT filters, RBAC route rules
|   |-- AuthTokenFilter.java                    # Bearer token JWT header extractor & SecurityContext builder
|   |-- TokenService.java                       # JWT cryptographic signing & token parsing utility
|   |-- CorsConfig.java                         # Global Cross-Origin Resource Sharing policy
|   |-- DataInitializer.java                    # Automated database seeder (Curriculum, Users, Locations)
|
|-- controller/                                 # REST API CONTROLLER ENDPOINTS
|   |-- AuthController.java                     # /api/auth/register, /api/auth/login, /api/auth/verify
|   |-- UserController.java                     # /api/users/profile, /api/users/{id}/saved-questions
|   |-- StudentAttendanceController.java        # /api/student/attendance/verify-location, /mark-lecture
|   |-- AttendanceController.java               # /api/attendance/class/**, /api/attendance/crt/**
|   |-- ManagementController.java               # /api/management/location, /api/management/stats
|   |-- SelectionWorkflowController.java        # /api/workflow/pipeline, /api/workflow/advance-round
|   |-- ResumeAtsController.java                # /api/resume-ats/analyze, /api/resume-ats/sample-jds
|   |-- HodMonitoringController.java            # /api/hod/faculty-performance, /api/hod/alerts
|   |-- CompanyController.java & DriveController.java # /api/companies, /api/drives
|
|-- model/                                      # JPA PERSISTENCE ENTITIES (MySQL Tables)
|   |-- User.java, Subject.java, StudentSubjectEnrollment.java, ClassAttendanceSession.java,
|   |-- ClassAttendanceRecord.java, InstituteLocation.java, Company.java, PlacementDrive.java...
|
|-- service/                                    # CORE BUSINESS LOGIC ENGINES
    |-- AttendanceService.java                  # Haversine distance calc, duplicate check, stats update
    |-- UserService.java                        # Multi-user profile persistence, auto-create student record
    |-- InstituteLocationService.java           # Geofence coordinates retrieval and audit trail
    |-- SelectionWorkflowService.java           # Round advancement, candidate scoring & offer emission
    |-- ResumeAtsService.java                   # Keyword tokenization, TF-IDF scoring, JD gap analysis
```

# 3. High-Level Multi-Tier Architectural Blueprint

PlacePrep enforces a decoupled Multi-Tier Architecture ensuring horizontal scalability, data consistency, and sub-100ms response latencies across both cloud and campus local network environments:

## Key Architectural Invariants & Design Principles:

- **Stateless Authentication Model:** The Spring Boot backend issues signed JSON Web Tokens (JWT) upon authentication. The client stores the bearer token securely and provides it via the Authorization: Bearer <token> header. The server holds zero HTTP session state, allowing horizontal scaling across instances.
- **Server-Side Authoritative Geofence Validation:** Although the browser computes distance for immediate UI feedback, **the server performs the definitive mathematical calculation** using configured campus coordinates in MySQL. Students cannot spoof attendance by manipulating client-side state.
- **Database-Level Unique Constraints:** Duplicate records are prohibited by composite unique constraints in MySQL (e.g. uk_session_student on (sessionId, studentId)). Even concurrent network retries are safely neutralized by the database layer.
- **User-Isolated LocalStorage Persistence:** All client caches are partitioned by normalized user email (placeprep_student_full_profile_{email}). Stale profile information from one user cannot bleed into another student's session.

# 4. Core Operational Workflows & Sequence Pipelines

## 4.1 Mandatory Campus GPS Geofencing Attendance Workflow (100m Rule & Face Match)

The Geofencing Attendance subsystem guarantees physical presence in classroom lectures. A student cannot mark attendance from a remote hostel, home, or proxy location:

```
1. SESSION DISCOVERY:
   Faculty schedules a class lecture session for a Subject (e.g. CS306 Cloud Computing).
   Status is 'OPEN'. The session appears in the student's Enrolled Subjects table and Attendance Widget.

2. BROWSER GPS CAPTURE:
   Student clicks 'Mark Attendance'. Frontend acquires high-accuracy GPS coordinates via HTML5 Geolocation API:
   navigator.geolocation.getCurrentPosition({ enableHighAccuracy: true, timeout: 10000, maximumAge: 0 })
   Student Latitude, Longitude, and GPS Accuracy (meters) are recorded.

3. HAVERSINE GEOFENCE VALIDATION:
   Client submits coordinates to: POST /api/student/attendance/verify-location
   Backend loads the active InstituteLocation from MySQL (e.g., Lat: 28.7499, Lon: 77.1170).
   Distance is computed via the spherical Haversine formula:
     a = sin²(Δϕ/2) + cos(ϕ1) * cos(ϕ2) * sin²(Δλ/2)
     c = 2 * atan2(√a, √(1-a))
     distanceMeters = Earth_Radius * c  (where Earth_Radius = 6,371,000 meters)
   If distanceMeters <= 100.0: Server returns { withinGeofence: true, distanceMeters, maxRadiusMeters: 100.0 }.
   If distanceMeters > 100.0:  Server rejects request with 400 Bad Request and detailed distance error.

4. WEBCAM LIVE PHOTO CAPTURE & FACIAL MATCH:
   Upon geofence validation, webcam stream opens via navigator.mediaDevices.getUserMedia({ video: true }).
   Student captures live selfie snapshot, verified against their registered profile photo in the database.

5. DATABASE PERSISTENCE & RECALCULATION:
   Client dispatches: POST /api/student/attendance/mark-lecture with:
   { sessionId, subjectCode, latitude, longitude, gpsAccuracy, faceVerified: true, studentEmail }
   Backend validates session is 'OPEN' and checks unique constraint uk_session_student.
   ClassAttendanceRecord is saved to MySQL with verificationMethod = 'GEOFENCE_AND_FACE'.
   Attended lecture count and percentage are recomputed in real time; Admit Card status updates.
```

## 4.2 Multi-User Profile Persistence & Relational Sync Workflow

A critical problem previously encountered was default profile fallback (e.g. Priyansh Jain profile appearing when logged in as Payanshi Jain). The upgraded architecture completely resolves this via two-way relational synchronization:

```
1. DYNAMIC STORAGE KEY PARTITIONING:
   When user logs in (e.g. Payanshi Jain, email: payanshijain10@gmail.com):
   The frontend derives an email-isolated storage key:
   storageKey = 'placeprep_student_full_profile_' + email.toLowerCase().replace(/[^a-z0-9]/g, '_')
   Stale profiles belonging to other student accounts are completely isolated and ignored.

2. ASYNCHRONOUS DATABASE FETCH:
   When opening UserProfileModal, client requests: GET /api/users/profile?email=payanshijain10@gmail.com
   If 'fullProfileJson' column exists in MySQL for this user, it is parsed and loaded into state.
   If no profile JSON exists yet, backend constructs an initial profile matching the user's
   registered credentials (name: 'Payanshi Jain', email: 'payanshijain10@gmail.com', roll: '23EJCAD122', batch: '2027').

3. RELATIONAL TWO-WAY MUTATION & COMMIT:
   When student updates any of the 11 profile sections and clicks 'Save Basic Details' or 'Save All':
   a) Client writes immediately to isolated LocalStorage.
   b) Client invokes: PUT /api/users/profile with { email, profile }
   c) The Spring Boot UserService:
      - Saves the serialized JSON payload to 'users.fullProfileJson' column in MySQL.
      - Automatically syncs top-level relational columns on the 'users' table:
        user.setName(basicDetails.fullName); user.setPhone(basicDetails.phone);
        user.setCollege(educationDetails.college); user.setBranch(educationDetails.branch);
        user.setRollNumber(educationDetails.rollNumber); user.setGraduationBatch(educationDetails.graduationBatch);
      - Emits a 'profile-updated' system event across frontend listeners.
```

## 4.3 Academic Roster & Class Period Session Management Workflow

- **Cascading Class Selectors:** Faculty selects Department (CSE, IT, ECE), Academic Year (1st through 4th), Semester (1 to 8), Subject (CS301 to CS310), Section (A, B, C), and Class Period (Period 1 09:00 AM to Period 6 04:00 PM).
- **Speed Attendance Marking:** Faculty clicks Present (P) or Absent (A) with one-click hotkeys. 'Mark All Present' speeds up registration for full attendance classes.
- **Shortage Radar (75% UGC/AICTE Mandate):** The system computes cumulative subject attendance. If a student's percentage drops below 75.0%, an instant red badge **Shortage (<75%)** is rendered and their End-Semester Exam Admit Card eligibility is revoked.
- **Institutional CSV/Excel Export:** One-click CSV exporter compiles all student roll numbers, names, attended counts, held counts, and shortage alerts for department meetings.

## 4.4 Placement Drives, Eligibility Radar & Multi-Round Selection Pipeline

- **Company Profile & Drive Creation:** TPO registers recruiters (e.g. Google, Microsoft, Amazon, Atlassian) specifying CTC, job description, target batch, and drive dates.
- **Automated Eligibility Radar:** Student criteria are validated by EligibilityService.java against minimum CGPA (e.g. 7.5), maximum allowed active backlogs (0 or 1), 10th/12th percentages, and eligible engineering departments. Ineligible students are blocked from applying with clear explanatory feedback.
- **Multi-Round Pipeline Stages:** Each drive defines consecutive rounds: Round 1: Online Assessment (OA) → Round 2: Technical Interview 1 → Round 3: System Design & Algorithms → Round 4: HR & Behavioral.
- **Candidate Advancement:** TPO/Recruiter reviews candidate evaluations, scores technical depth, problem-solving, and communication on a scale of 1-100, and advances candidates to the next round.
- **Offer Issuance & Acceptance:** Final round qualifiers receive digital offer status with designation, stipend, and joining date recorded in the selection_results database table.

## 4.5 In-Browser DSA Studio & AI ATS Resume Scanner

- **DSA Practice Studio:** Curated library of 350+ data structures and algorithm challenges mapped by company tags (FAANG, Tier-1) and topic categories (Dynamic Programming, Graphs, Tries, Sliding Window). Includes built-in JavaScript/Python execution sandbox that validates edge cases.
- **AI ATS Resume Analyzer:** Ingests candidate resumes, parses core sections (Experience, Education, Skills, Projects, Achievements), compares keywords against real-world Job Descriptions, and scores ATS readability, keyword density, and structural completeness out of 100.

# 5. Relational Database Schema & Entity Relationship Specification

The backend leverages a normalized MySQL 8.0 relational schema mapped via Hibernate/JPA entities. The following tables represent the foundational storage model:

## Table: users

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| `id` | `BIGINT` | PRIMARY KEY, AUTO_INCREMENT |
| `name` | `VARCHAR(255)` | NOT NULL - Full Name of User |
| `email` | `VARCHAR(255)` | NOT NULL, UNIQUE - User Email Key |
| `password` | `VARCHAR(255)` | NOT NULL - BCrypt Hashed Password |
| `role` | `VARCHAR(50)` | NOT NULL - student, teacher, hod, tpo, management |
| `status` | `VARCHAR(50)` | NOT NULL - ACTIVE, SUSPENDED |
| `roll_number` | `VARCHAR(100)` | Student University Roll Number (e.g. 23EJCAD122) |
| `college` | `VARCHAR(255)` | Institutional Affiliation (e.g. DTU) |
| `branch` | `VARCHAR(255)` | Academic Branch (e.g. Computer Science and Engineering) |
| `graduation_batch` | `VARCHAR(50)` | Graduation Year (e.g. 2027) |
| `phone` | `VARCHAR(50)` | Contact Mobile Number |
| `cgpa` | `DOUBLE` | Cumulative Grade Point Average (Default 8.5) |
| `tenth_percentage` | `DOUBLE` | Secondary School Certificate Percentage |
| `twelfth_percentage` | `DOUBLE` | Higher Secondary School Certificate Percentage |
| `active_backlogs` | `INT` | Count of Active Academic Backlogs |
| `full_profile_json` | `LONGTEXT` | Complete 11-section serialized student JSON dossier |

## Table: subjects

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| `id` | `BIGINT` | PRIMARY KEY, AUTO_INCREMENT |
| `code` | `VARCHAR(50)` | NOT NULL, UNIQUE - Subject Code (e.g. CS306) |
| `name` | `VARCHAR(255)` | NOT NULL - Full Subject Name |
| `department` | `VARCHAR(100)` | NOT NULL - Department (e.g. CSE) |
| `semester` | `VARCHAR(20)` | NOT NULL - Academic Semester (e.g. 6) |
| `faculty_name` | `VARCHAR(255)` | Assigned Instructor Name |
| `faculty_email` | `VARCHAR(255)` | Assigned Instructor Email |
| `credits` | `INT` | Academic Credit Units (Default 4) |
| `min_attendance_percentage` | `DOUBLE` | Minimum Mandate (Default 75.0%) |

## Table: class_attendance_sessions

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| id | BIGINT | PRIMARY KEY, AUTO_INCREMENT |
| subject_code | VARCHAR(50) | NOT NULL - Foreign reference to subjects.code |
| lecture_number | INT | Lecture Index in Syllabus (e.g. Lecture 1, 2, 3) |
| lecture_title | VARCHAR(255) | Topic Covered (e.g. Virtualization & Containers) |
| session_date | VARCHAR(50) | NOT NULL - Date string (YYYY-MM-DD) |
| start_time | VARCHAR(50) | Lecture Start Time (e.g. 09:00 AM) |
| end_time | VARCHAR(50) | Lecture End Time (e.g. 10:00 AM) |
| status | VARCHAR(50) | NOT NULL - OPEN, CLOSED, CANCELLED |
| institute_name | VARCHAR(255) | Associated Institution |

## Table: class_attendance_records

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| id | BIGINT | PRIMARY KEY, AUTO_INCREMENT |
| session_id | BIGINT | NOT NULL - FK to class_attendance_sessions.id |
| student_id | BIGINT | NOT NULL - FK to users.id |
| student_name | VARCHAR(255) | Cached Student Name |
| roll_number | VARCHAR(100) | Cached Student University Roll |
| status | VARCHAR(50) | NOT NULL - PRESENT, ABSENT, LATE |
| distance_meters | DOUBLE | Haversine calculated distance from campus |
| gps_accuracy | DOUBLE | Reported browser GPS accuracy in meters |
| verification_method | VARCHAR(100) | GEOFENCE_AND_FACE, MANUAL_FACULTY |
| marked_by | VARCHAR(255) | Principal or Faculty identifier who confirmed attendance |
| marked_at | VARCHAR(100) | Timestamp of attendance submission |
| UNIQUE CONSTRAINT | uk_session_student | UNIQUE KEY (session_id, student_id) |

## Table: institute_locations

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| id | BIGINT | PRIMARY KEY, AUTO_INCREMENT |
| institute_name | VARCHAR(255) | NOT NULL - Official Institute Name |
| latitude | DOUBLE | NOT NULL - Authoritative Campus Latitude (e.g. 28.7499) |
| longitude | DOUBLE | NOT NULL - Authoritative Campus Longitude (e.g. 77.1170) |
| radius_meters | DOUBLE | NOT NULL - Fixed Geofence Boundary (100.0 meters) |
| active | BOOLEAN | NOT NULL - Active verification baseline flag |
| updated_by | VARCHAR(255) | Management user who last configured coordinates |
| updated_at | VARCHAR(100) | Audit timestamp of coordinate modification |

## Table: placement_drives & candidates

| Column Name | SQL Data Type | Constraints & Semantic Description |
|---|---|---|
| **drive_id** | BIGINT | PRIMARY KEY, AUTO_INCREMENT |
| **company_name** | VARCHAR(255) | Recruiting Organization (e.g. Microsoft, Google) |
| **role_title** | VARCHAR(255) | Job Role (e.g. Software Development Engineer) |
| **ctc_lpa** | DOUBLE | Annual Compensation Package in Lakhs Per Annum |
| **min_cgpa** | DOUBLE | Mandatory CGPA Cutoff |
| **max_backlogs** | INT | Maximum Allowed Backlogs |
| **drive_status** | VARCHAR(50) | UPCOMING, ACTIVE, COMPLETED |
| **candidate_id** | VARCHAR(100) | Unique Candidate Code (e.g. PP-5847994) |
| **current_round** | VARCHAR(100) | Current Selection Stage (OA, Tech 1, Tech 2, HR) |
| **selection_status** | VARCHAR(50) | APPLIED, SHORTLISTED, SELECTED, REJECTED |

# 6. Complete REST API Endpoint Catalog & Permissions Matrix

All client-server communications operate over standard HTTP/1.1 and HTTP/2 JSON REST contracts. Endpoints are protected by Spring Security role matchers as detailed below:

| Method | Endpoint URI | Security Role | Functional Purpose |
|---|---|---|---|
| POST | /api/auth/register | PermitAll | Register user with role, password, and credentials |
| POST | /api/auth/login | PermitAll | Validate email/password and issue signed JWT token |
| GET | /api/users/profile | PermitAll / Auth | Retrieve 11-section profile by email or JWT principal |
| PUT | /api/users/profile | PermitAll / Auth | Save student profile JSON with users MySQL table |
| PUT | /api/users/{id}/batch | Authenticated | Update student graduation batch assignment |
| GET | /api/users/{id}/saved-questions | Authenticated | Retrieve student's bookmarked practice questions |
| POST | /api/student/attendance/verify-location | PermitAll | Validate GPS coordinates against active campus (100m) |
| POST | /api/student/attendance/verify-face | PermitAll | Compare captured selfie against registered photo |
| POST | /api/student/attendance/mark-lecture | PermitAll / Student | Save verified lecture attendance record to MySQL |
| GET | /api/student/attendance/subjects | PermitAll / Student | Get student enrolled subjects with percentages |
| GET | /api/student/attendance/subjects/{code}/lectures | PermitAll | Get scheduled lecture sessions and history for subject |
| GET | /api/attendance/class/roster | TEACHER, HOD, ADMIN | Fetch class student roster for speed attendance marking |
| POST | /api/attendance/class/mark | TEACHER, HOD, ADMIN | Save bulk faculty class attendance records |
| GET | /api/attendance/class/report | TEACHER, HOD, ADMIN | Fetch cumulative semester attendance shortage report |
| GET | /api/attendance/crt/sessions | PermitAll | List scheduled Campus Recruitment Training sessions |
| POST | /api/attendance/crt/sessions | TEACHER, TPO, ADMIN | Schedule and create new CRT training module |
| POST | /api/attendance/crt/sessions/{id}/mark | TEACHER, TPO, ADMIN | Submit faculty CRT session attendance records |
| GET | /api/management/location | MANAGEMENT, ADMIN | Fetch configured college GPS geofence coordinates |
| PUT | /api/management/location | MANAGEMENT, ADMIN | Update campus latitude, longitude, and 100m radius |
| GET | /api/companies | PermitAll | List registered recruitment companies & past drives |
| POST | /api/companies | ADMIN, TPO | Register new corporate recruitment partner |
| GET | /api/drives | PermitAll | List active and upcoming campus recruitment drives |
| POST | /api/drives | ADMIN, TPO | Create new placement drive with eligibility rules |
| GET | /api/workflow/pipeline | Authenticated | Fetch candidate progression pipeline |
| POST | /api/workflow/advance-round | ADMIN, TPO | Promote shortlisted candidates to next interview stage |
| POST | /api/resume-ats/analyze | PermitAll | Analyze uploaded resume against target job description |
| GET | /api/hod/faculty-performance | HOD, ADMIN | Review faculty class conduction and attendance audits |
| GET | /api/academic-records/student/{id} | Authenticated | Retrieve marksheet, semester SGPAs and CGPA |

# 7. Deployment Topology, Runtime Configuration & Data Integrity

PlacePrep is engineered for seamless cloud native deployments with enterprise failover resilience. The following specifications detail the production runtime topology:

| Deployment Dimension | Production Infrastructure Specification |
|---|---|
| **Frontend Hosting (Vercel)** | Deployed as a high-speed Single Page Application on Vercel's global Edge CDN (URL: https://place-prep-alpha.vercel.app/). Serves static assets over HTTP/3 with automatic SSL/TLS termination, Brotli compression, and immutable cache-control headers. vercel.json enforces SPA rewrites ensuring deep client-side routes resolve cleanly without 404 errors. |
| **Backend Microservice (Spring Boot)** | Packaged as an executable JAR running on a dedicated Linux / Windows cloud host on port 8082 with Java 23 LTS runtime. Embedded Apache Tomcat web server configured with non-blocking NIO connectors, capable of servicing 5,000+ concurrent student attendance submissions during peak class changeover windows. |
| **Database Layer (MySQL 8.0)** | Hosted on MySQL 8.0 on port 3306 utilizing the ACID-compliant InnoDB storage engine. Managed by HikariCP connection pooling configured with minimum-idle: 5, maximum-pool-size: 20, and idle-timeout: 300000ms. Automated schema migration via Hibernate ddl-auto: update. |
| **HTTPS & Geolocation Compliance** | HTML5 Geolocation and WebRTC camera stream APIs strictly require a **Secure Context (HTTPS)** in modern browsers (Chrome, Edge, Safari). Production deployments mandate SSL certificates across both frontend and backend domains to guarantee location access without browser permission blocks. |
| **Fault-Tolerant Hybrid Fallback** | To prevent catastrophic failures during cloud network disruptions, client services (attendanceStorage.js, crtAttendanceStorage.js, storage.js) implement a resilient hybrid caching layer. If the backend is temporarily unreachable, critical roster views gracefully fall back to local storage and queue mutations for synchronization. |

**ARCHITECTURAL SIGN-OFF & CERTIFICATION:**
This technical specification certifies that the PlacePrep software architecture complies with modern enterprise software engineering standards, including strict layer isolation, cryptographic password hashing, authoritative server-side geofencing verification, relational database normalization, and robust cross-origin resource sharing policies.

**Document Reference:** PP-ARCH-2026-V2 | **Status:** Production Ready & Approved | **Approved By:** Lead Systems Architect